

Geometry-based Methods in Vision

CMU 16-822, Fall 2024

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Lecture 14b: Three-view Geometry

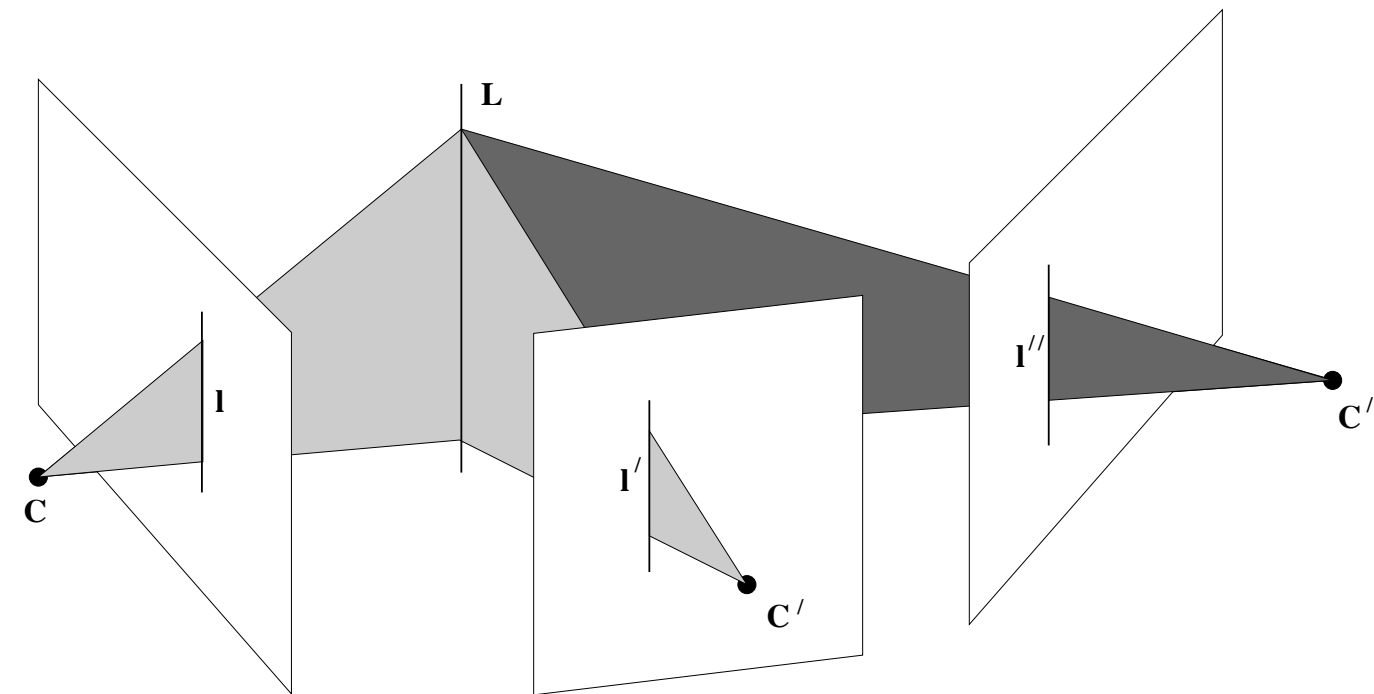


Three-view Geometry: Motivation

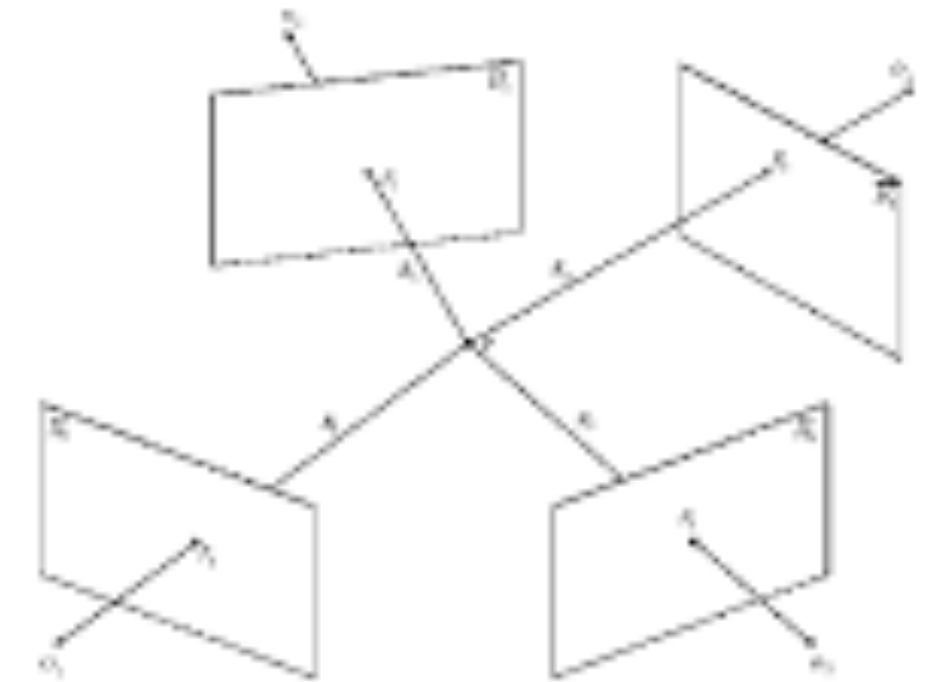
(if it's not in the assignments, why should I care?)



Real-world systems can have 3 camera setups

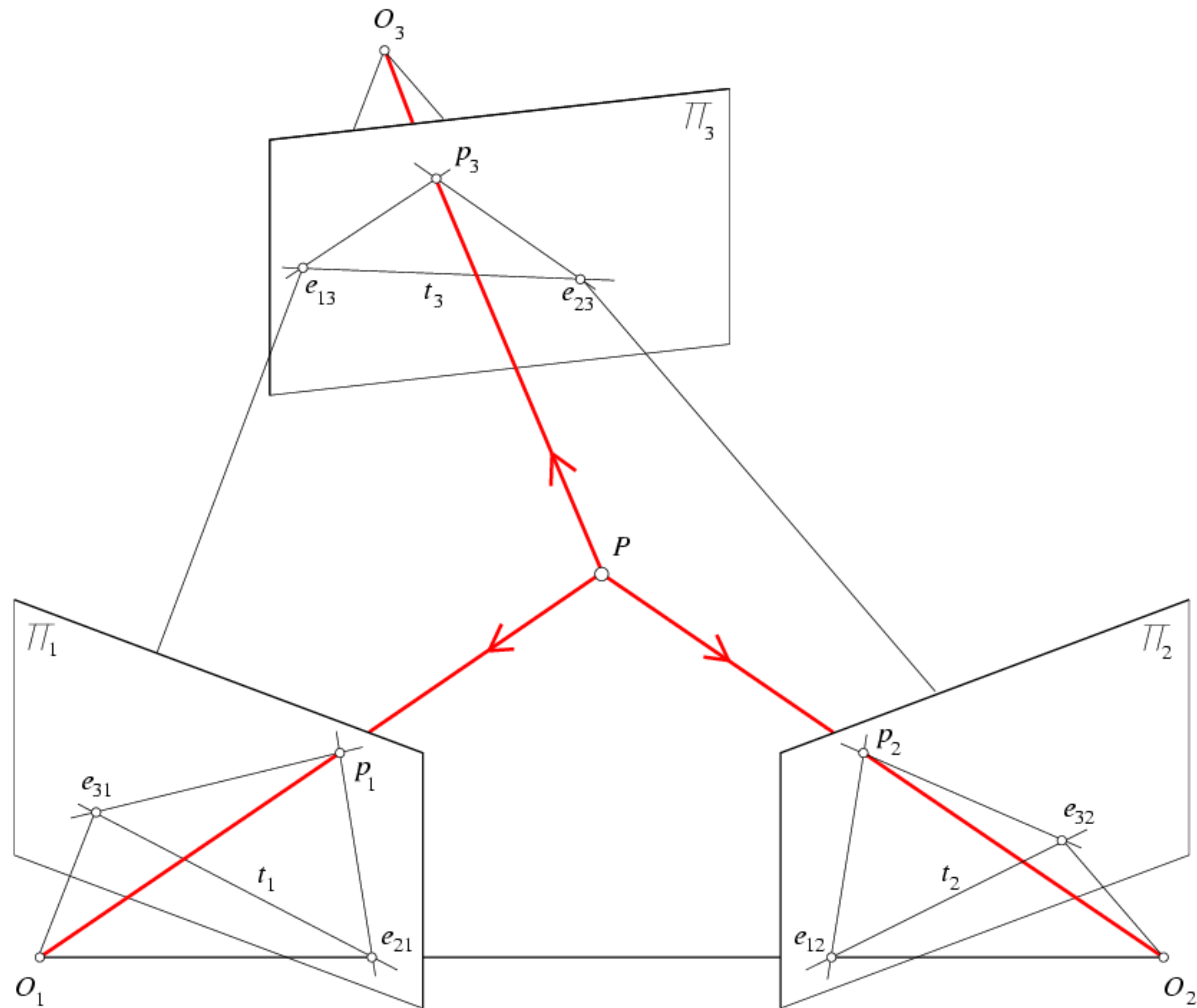


Core concepts can be helpful in unexpected ways



Helps appreciate the difficulty of analytically solving n-view setups!

Three-view Geometry: Setup



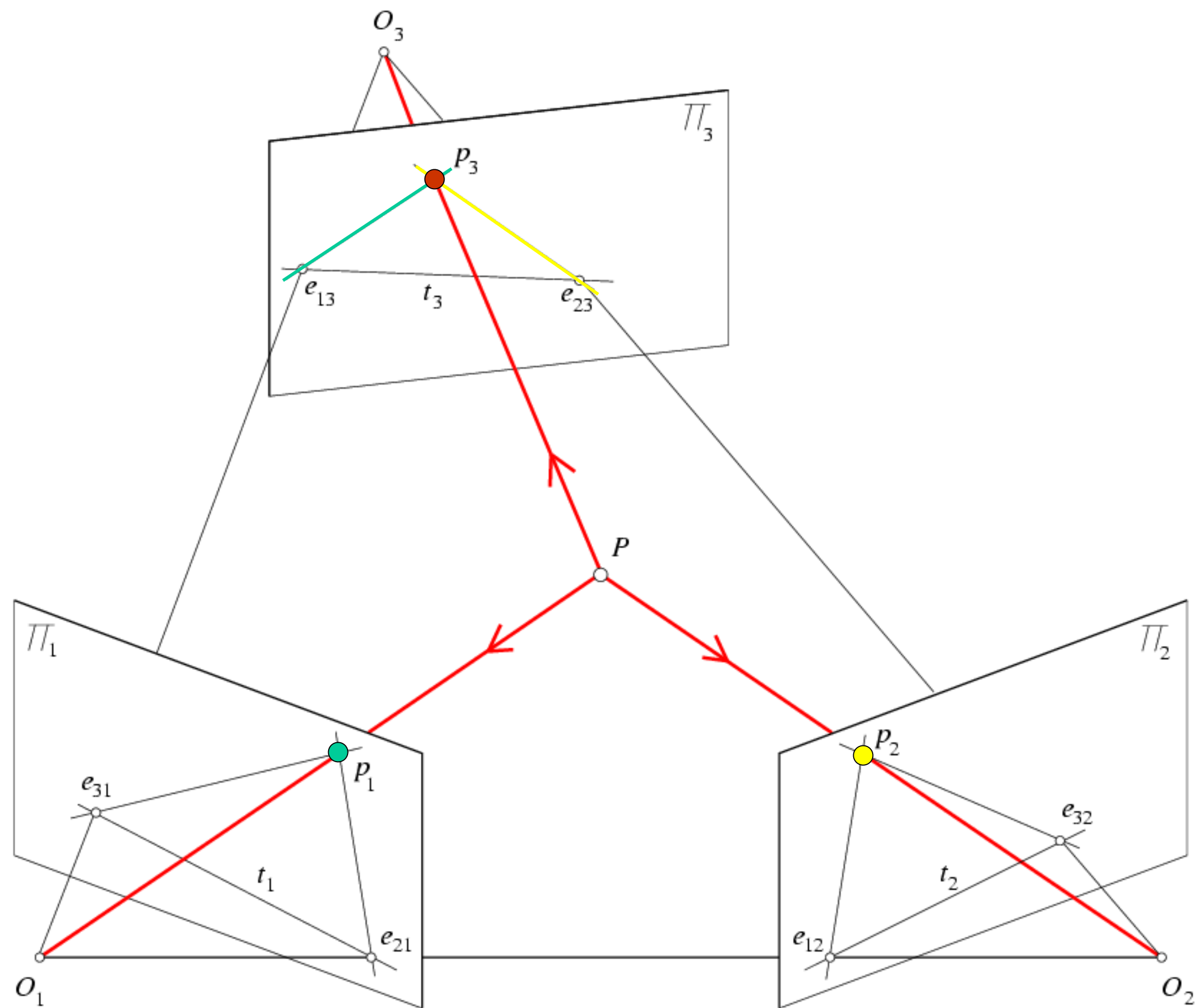
Camera Matrices:

$$(\mathbf{P}, \mathbf{P}', \mathbf{P}'')$$

$$F_{12}, F_{13}, F_{23},$$

Q: Do we need to develop more concepts than fundamental matrices? Or can perform all operations given these?

Finding Corresponding Projections

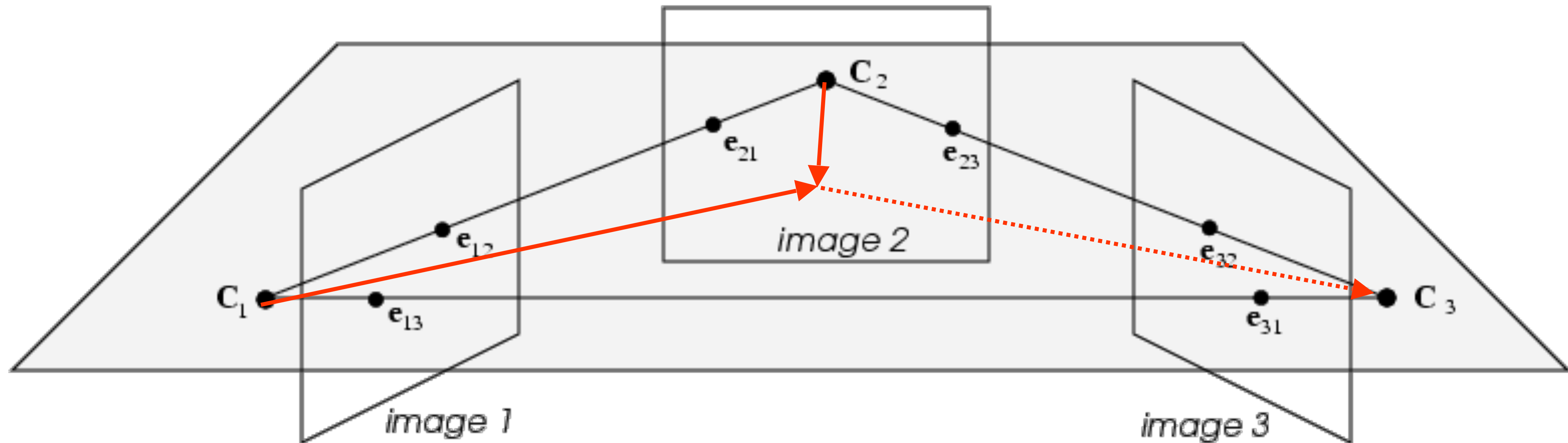


Given projections of a 3D point in two images (green, yellow), can we find its projection in the 3rd image?

Using fundamental matrices: Find intersection of the two epipolar lines!

But, is this always a valid approach?

Insufficiency of Pairwise Fundamental Matrices

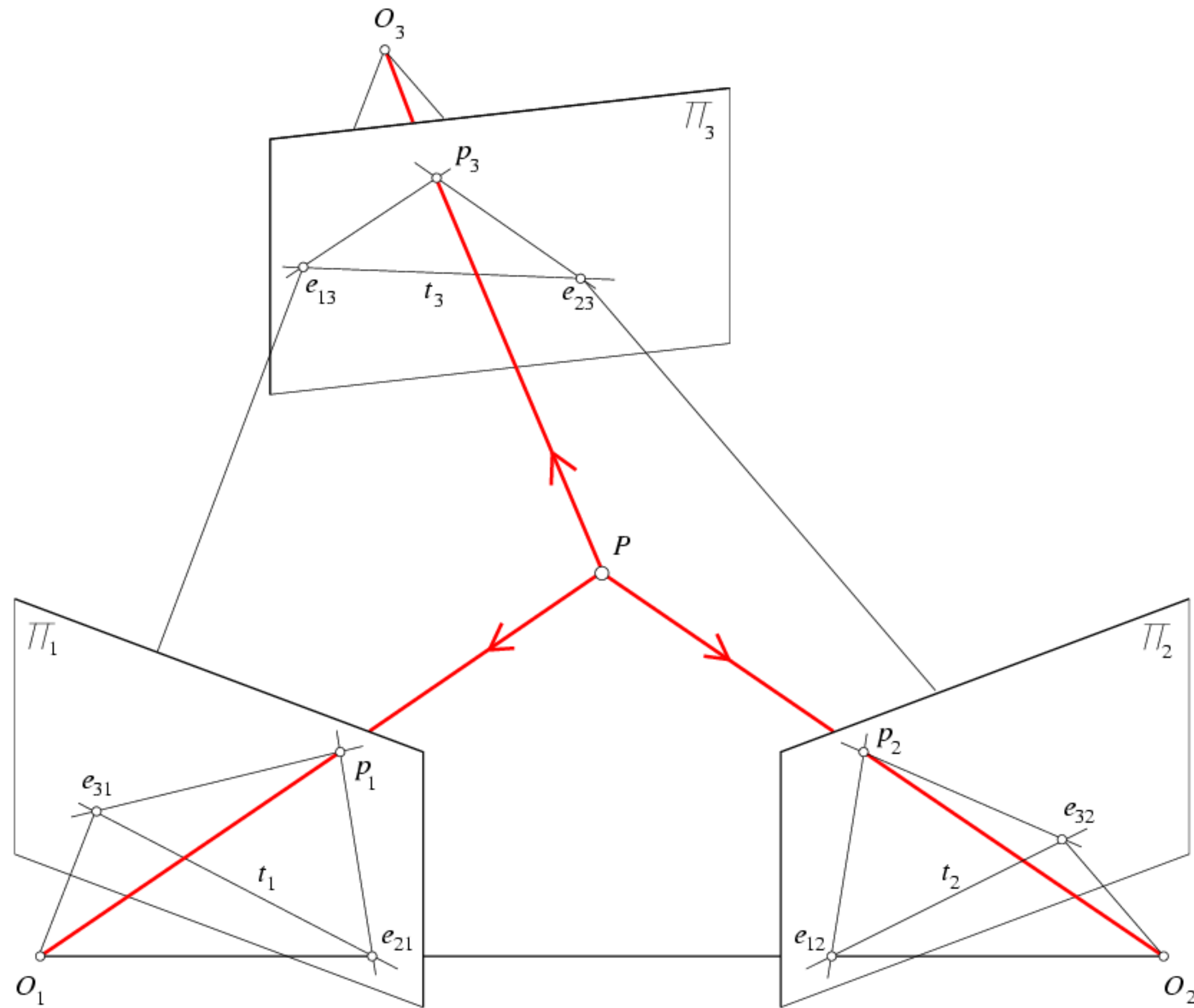


Using fundamental matrices: Find intersection of the two epipolar lines!

But, for 3D points in the plane containing camera centers, we do not get distinct epipolar lines from two images!

But, triangulation suggests identifying projection is still feasible!

Three-view Geometry: Setup



Camera Matrices:

$$(\mathbf{P}, \mathbf{P}', \mathbf{P}'')$$

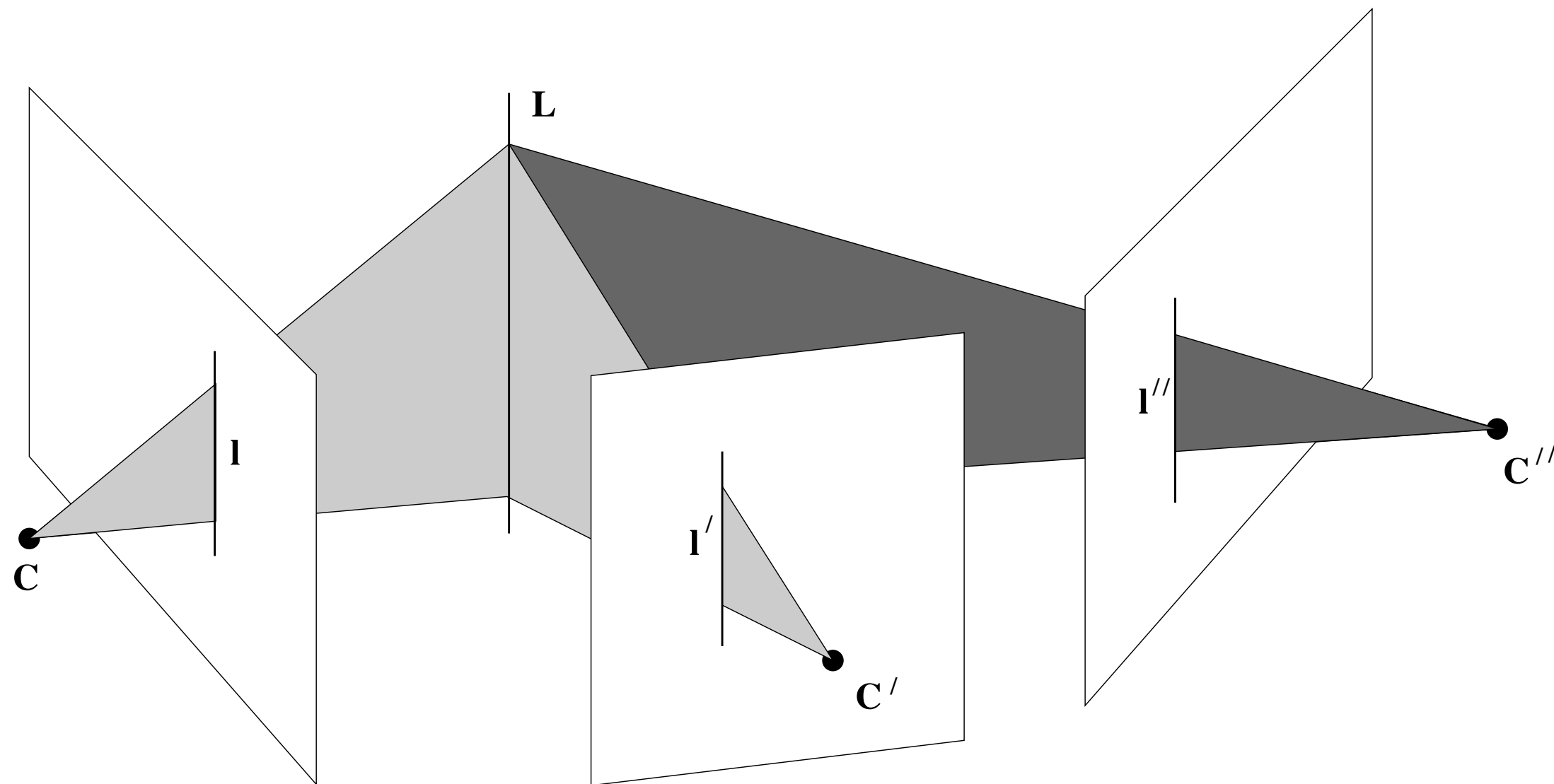
Say we have corresponding point projections:

$$(\mathbf{x}, \mathbf{x}', \mathbf{x}'')$$

Can we find one given other two using some structure other than fundamental matrices?

Yes, but will take us a while to get there!

2D Projections of a 3D Line



Say we have corresponding line projections
(i.e. 2D line projections of some 3D line):

$$(l, l', l'')$$

Q: What is the (3D) plane defined by l and C (given P)?

$$\pi = P^T l \quad \pi' = P'^T l' \quad \pi'' = P''^T l''$$

Let's consider the 4×3 matrix: $\begin{bmatrix} \pi & \pi' & \pi'' \end{bmatrix}$

Q: What is the rank of this matrix?

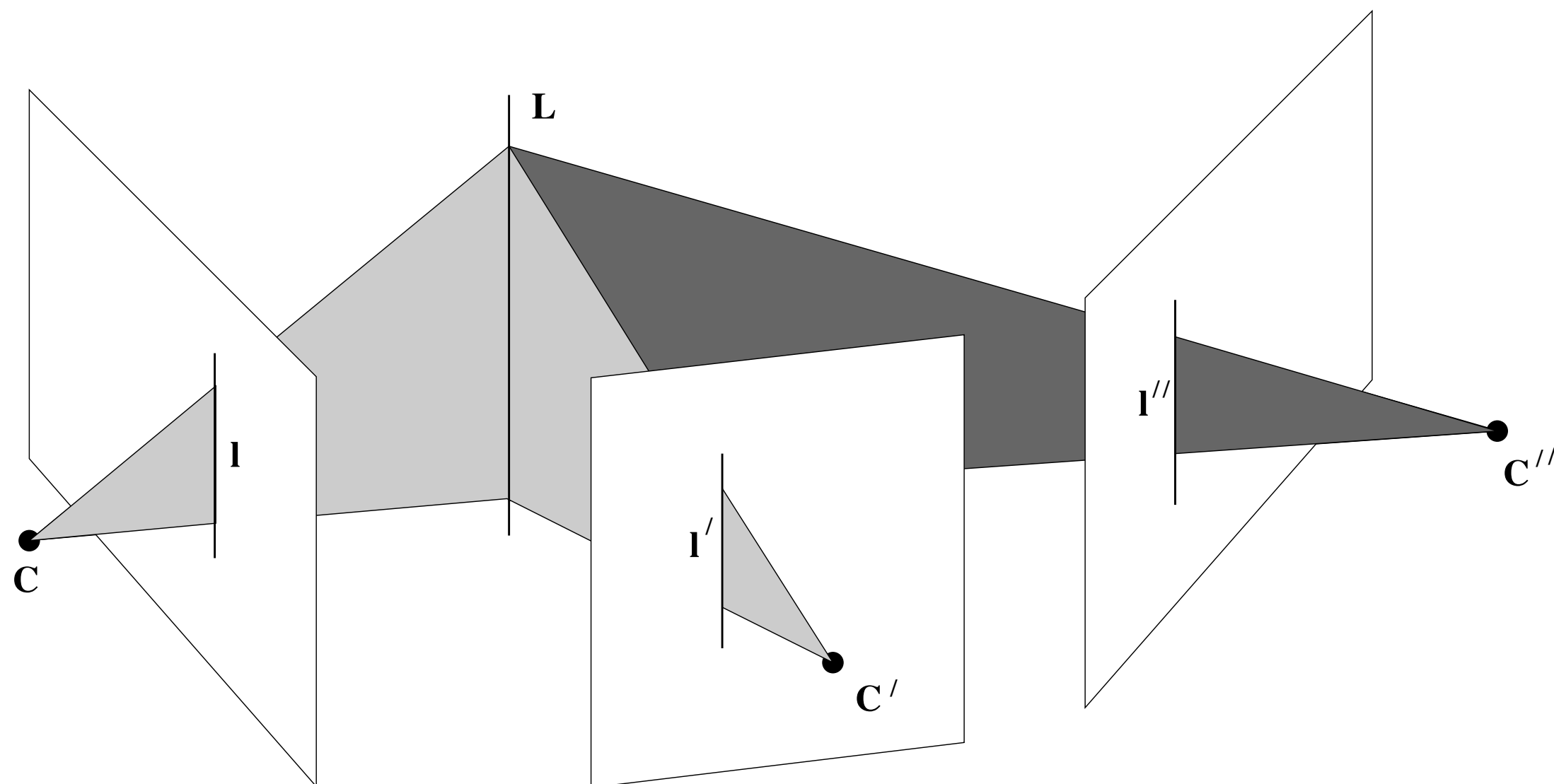
Hint: The intersection of these planes is a line

$$\pi = \alpha \pi' + \beta \pi''$$

If we know l' , l'' (and camera matrices)— we can find
 l , α , β

Q: How many unknowns? How many equations?

Relating corresponding lines: Trifocal Tensor



Say we have corresponding line projections
(i.e. 2D line projections of some 3D line):

$$(\mathbf{l}, \mathbf{l}', \mathbf{l}'')$$

$$\boldsymbol{\pi} = \mathbf{P}^T \mathbf{l} \quad \boldsymbol{\pi}' = \mathbf{P}'^T \mathbf{l}' \quad \boldsymbol{\pi}'' = \mathbf{P}''^T \mathbf{l}''$$

$$\boldsymbol{\pi} = \alpha \boldsymbol{\pi}' + \beta \boldsymbol{\pi}''$$

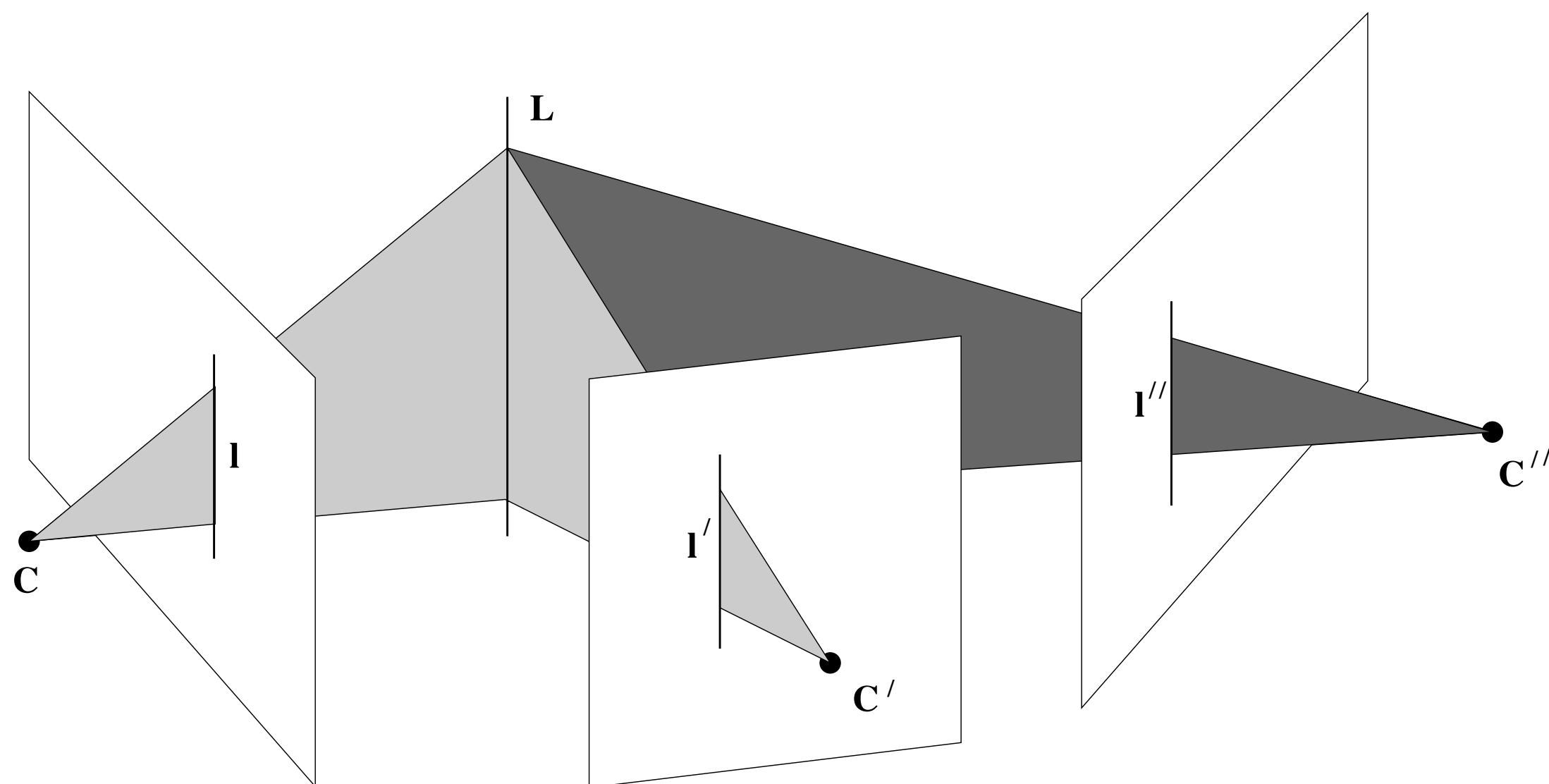
Can determine $\mathbf{l}$ given $\mathbf{l}', \mathbf{l}''$ —

$$\mathbf{l} \equiv (l_1, l_2, l_3) \quad l_i = \mathbf{l}'^T \mathbf{T}_i \mathbf{l}''$$

$$\mathbf{l}^T = \mathbf{l}'^T [\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3] \mathbf{l}''$$

Tri-focal Tensor
(3 matrices of size 3 X 3)

Relating corresponding lines: Trifocal Tensor



Say we have corresponding line projections
(i.e. 2D line projections of some 3D line):

$$(\mathbf{l}, \mathbf{l}', \mathbf{l}'')$$

$$\mathbf{l}^\top = \mathbf{l}'^\top [\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3] \mathbf{l}''$$

Can derive $\mathbf{T}$ in canonical camera form:

$$\mathbf{P} = [\mathbf{I} \mid \mathbf{0}] \quad \mathbf{P}' = [\mathbf{A} \mid \mathbf{a}_4] \quad \mathbf{P}'' = [\mathbf{B} \mid \mathbf{b}_4]$$

$$\boldsymbol{\pi} = \mathbf{P}^\top \mathbf{l} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \boldsymbol{\pi}' = \mathbf{P}'^\top \mathbf{l}' = \begin{pmatrix} \mathbf{A}^\top \mathbf{l}' \\ \mathbf{a}_4^\top \mathbf{l}' \end{pmatrix} \quad \boldsymbol{\pi}'' = \mathbf{P}''^\top \mathbf{l}'' = \begin{pmatrix} \mathbf{B}^\top \mathbf{l}'' \\ \mathbf{b}_4^\top \mathbf{l}'' \end{pmatrix}$$

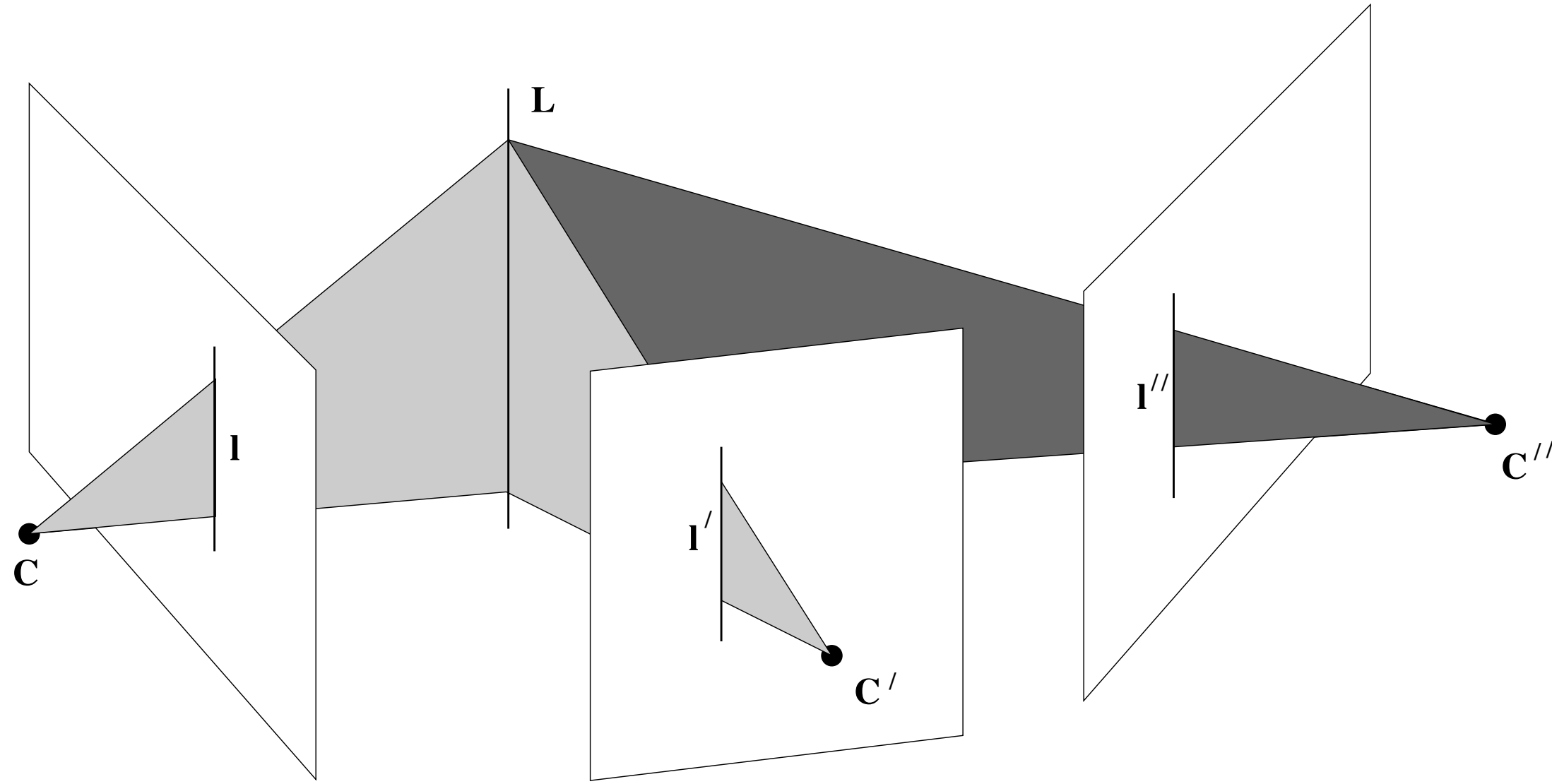
$$\boldsymbol{\pi} = \alpha \boldsymbol{\pi}' + \beta \boldsymbol{\pi}'' \quad \alpha = k(\mathbf{b}_4^\top \mathbf{l}'') \text{ and } \beta = -k(\mathbf{a}_4^\top \mathbf{l}')$$

$$\mathbf{l} = (\mathbf{b}_4^\top \mathbf{l}'') \mathbf{A}^\top \mathbf{l}' - (\mathbf{a}_4^\top \mathbf{l}') \mathbf{B}^\top \mathbf{l}''$$

$$\mathbf{T}_i = \mathbf{a}_i \mathbf{b}_4^\top - \mathbf{a}_4 \mathbf{b}_i^\top$$

(Just like $\mathbf{F}$ is invariant to 3D homographies, so is $\mathbf{T}$! So studying it's form/existence in a restricted case is still informative)

Trifocal Tensor



Say we have corresponding line projections
(i.e. 2D line projections of some 3D line):

$$(\mathbf{l}, \mathbf{l}', \mathbf{l}'')$$

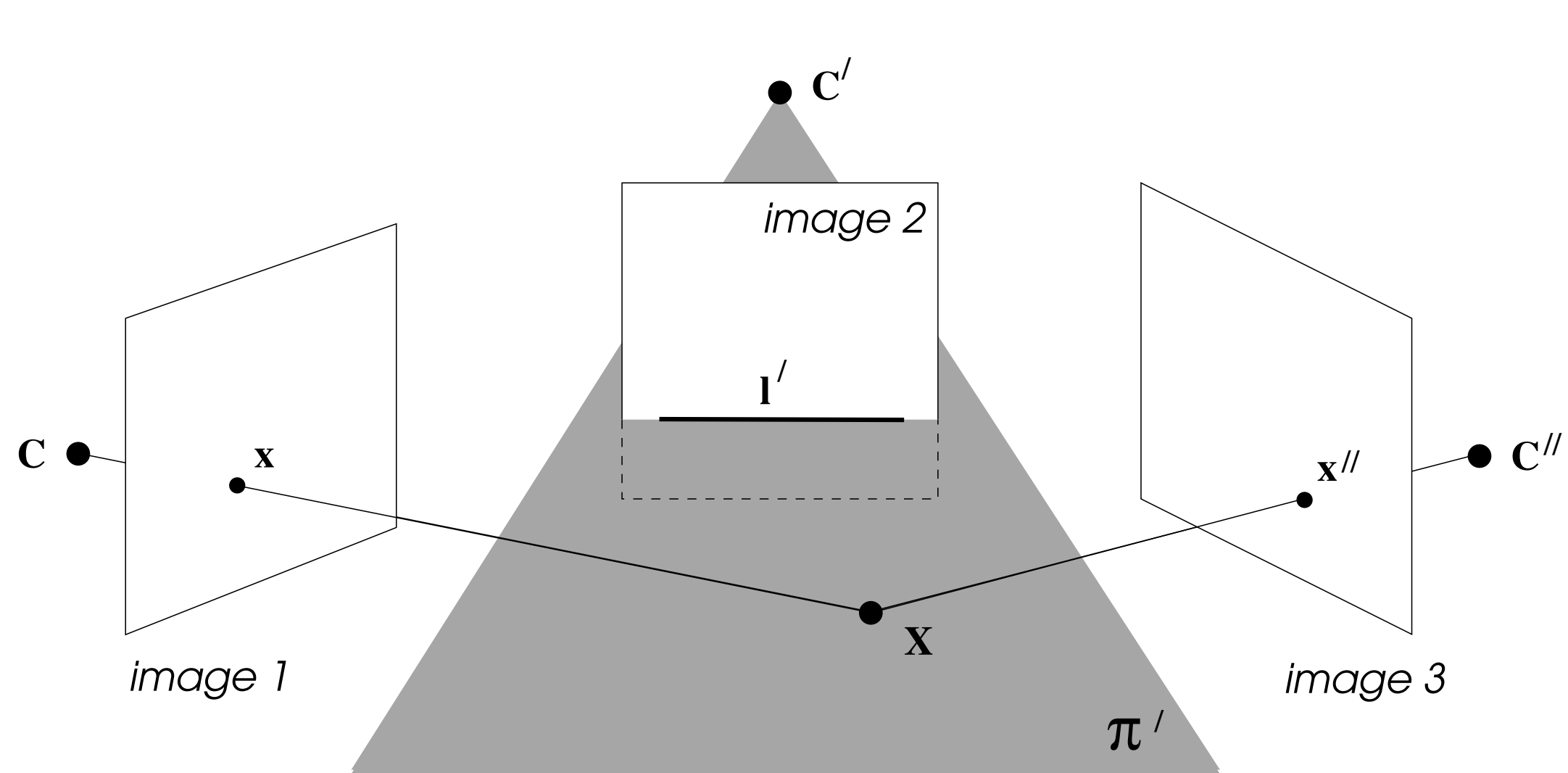
$$\mathbf{l}^T = \mathbf{l}'^T [T_1, T_2, T_3] \mathbf{l}''$$

Q: If $\mathbf{T}$ captures the configuration of 3 cameras upto a 3D homography (i.e. all cameras being transformed by $P \rightarrow PH$), how many DoFs should $\mathbf{T}$ have?

$$33 - 15 = 18$$

Given $\mathbf{T}$, can we express other relations between points and lines?

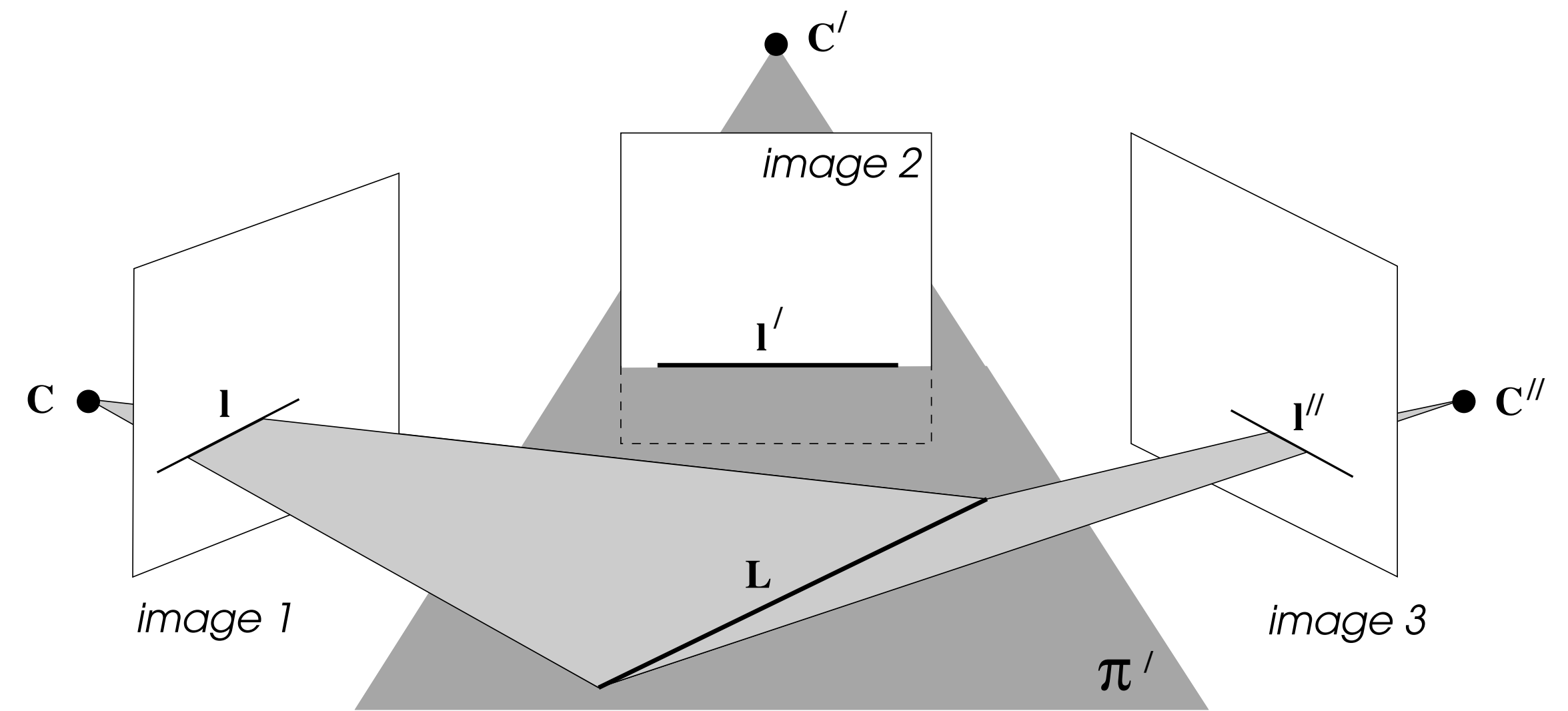
Trifocal Tensor: Inducing Planar Homographies



Let's consider a line $\mathbf{l}'$ and the plane $\boldsymbol{\pi}' = \mathbf{P}'^T \mathbf{l}'$

Projections via this plane induce some homography $\mathbf{H}$ between images 1 and 3:

$$\mathbf{x}'' = \mathbf{H}\mathbf{x} \quad \mathbf{l} = \mathbf{H}^T \mathbf{l}''$$

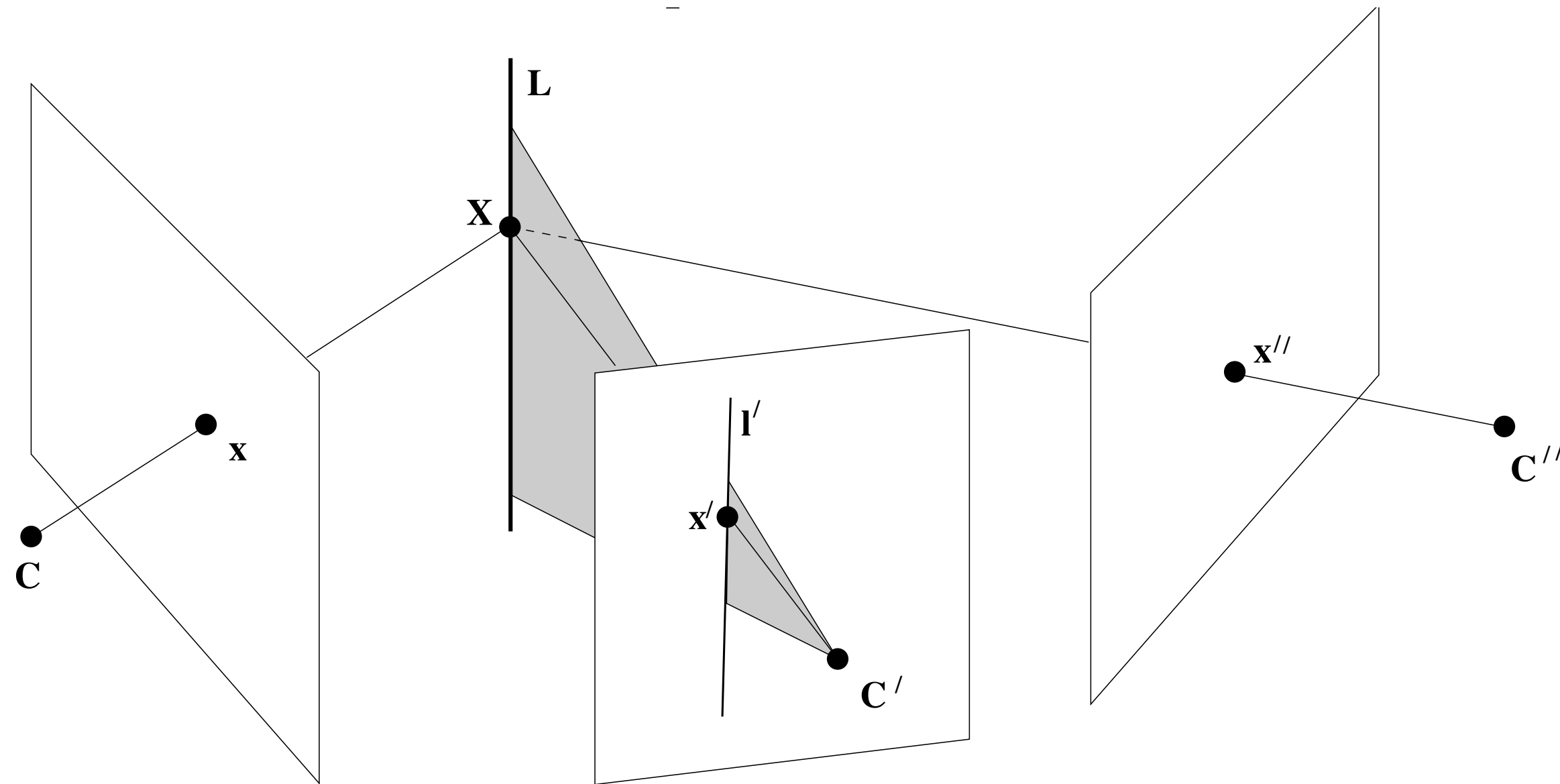


But, given a line $\mathbf{l}''$, we can compute $\mathbf{l}$ in a different way! $\mathbf{l}^T = \mathbf{l}'^T [\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3] \mathbf{l}''$

Thus, $\mathbf{T}$ lets us compute the homography induced by $\boldsymbol{\pi}' = \mathbf{P}'^T \mathbf{l}'$

$$\mathbf{H}_{13}(\mathbf{l}') = [\mathbf{T}_1^T, \mathbf{T}_2^T, \mathbf{T}_3^T] \mathbf{l}'$$

Trifocal Tensor: Relating Points and Lines



Let's consider a 3D point which we know projects to $(\mathbf{x}, \mathbf{x}'')$, and lies on line $\mathbf{l}'$

$$\mathbf{x}'' = H_{13}(\mathbf{l}') \mathbf{x}$$

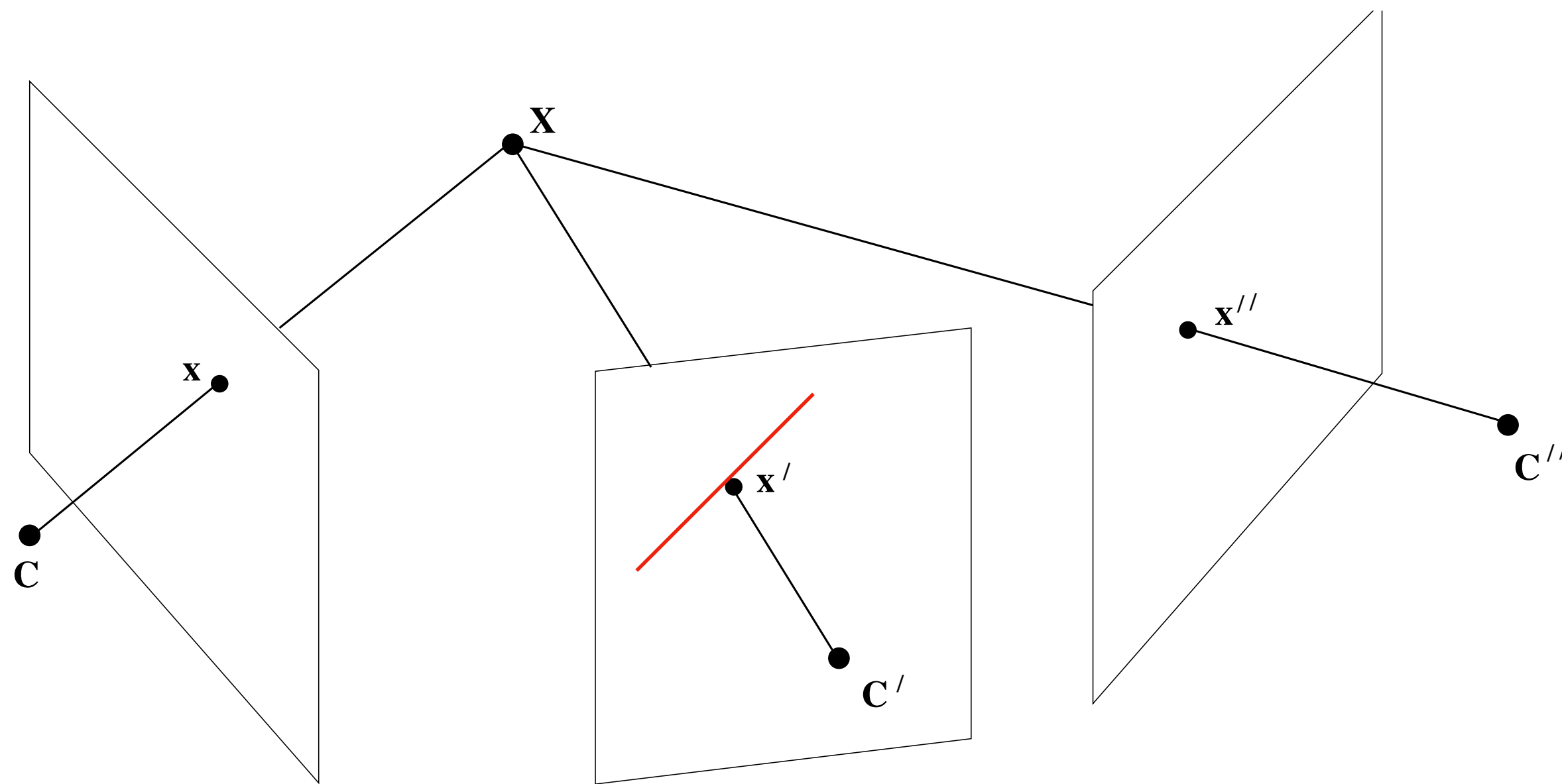
$$= [T_1^\top \mathbf{l}', T_2^\top \mathbf{l}', T_3^\top \mathbf{l}'] \mathbf{x}$$

$$= \left(\sum_i x^i T_i^\top \right) \mathbf{l}'$$

$$\mathbf{x}''^\top [\mathbf{x}'']_\times = \mathbf{0}^\top$$

$$\mathbf{l}'^\top \left(\sum_i x^i T_i \right) [\mathbf{x}'']_\times = \mathbf{0}^\top$$

Trifocal Tensor: Relating points projection



Let's (finally!) consider corresponding projections ($\mathbf{x}, \mathbf{x}', \mathbf{x}''$)

Can find one given the other two!

For **any** line $\mathbf{l}'$ passing $\mathbf{x}'$:

$$\mathbf{l}'^T \left(\sum_i x^i \mathbf{T}_i \right) [\mathbf{x}'']_{\times} = \mathbf{0}^T$$

Given **any** $\mathbf{y}'$, we can construct $\mathbf{l}'$ passing $\mathbf{x}'$ and $\mathbf{y}'$:

$$\mathbf{l}' = \mathbf{x}' \times \mathbf{y}' = [\mathbf{x}']_{\times} \mathbf{y}'$$

For **all** $\mathbf{y}'$:

$$\mathbf{y}'^T [\mathbf{x}']_{\times} \left(\sum_i x^i \mathbf{T}_i \right) [\mathbf{x}'']_{\times} = \mathbf{0}^T$$

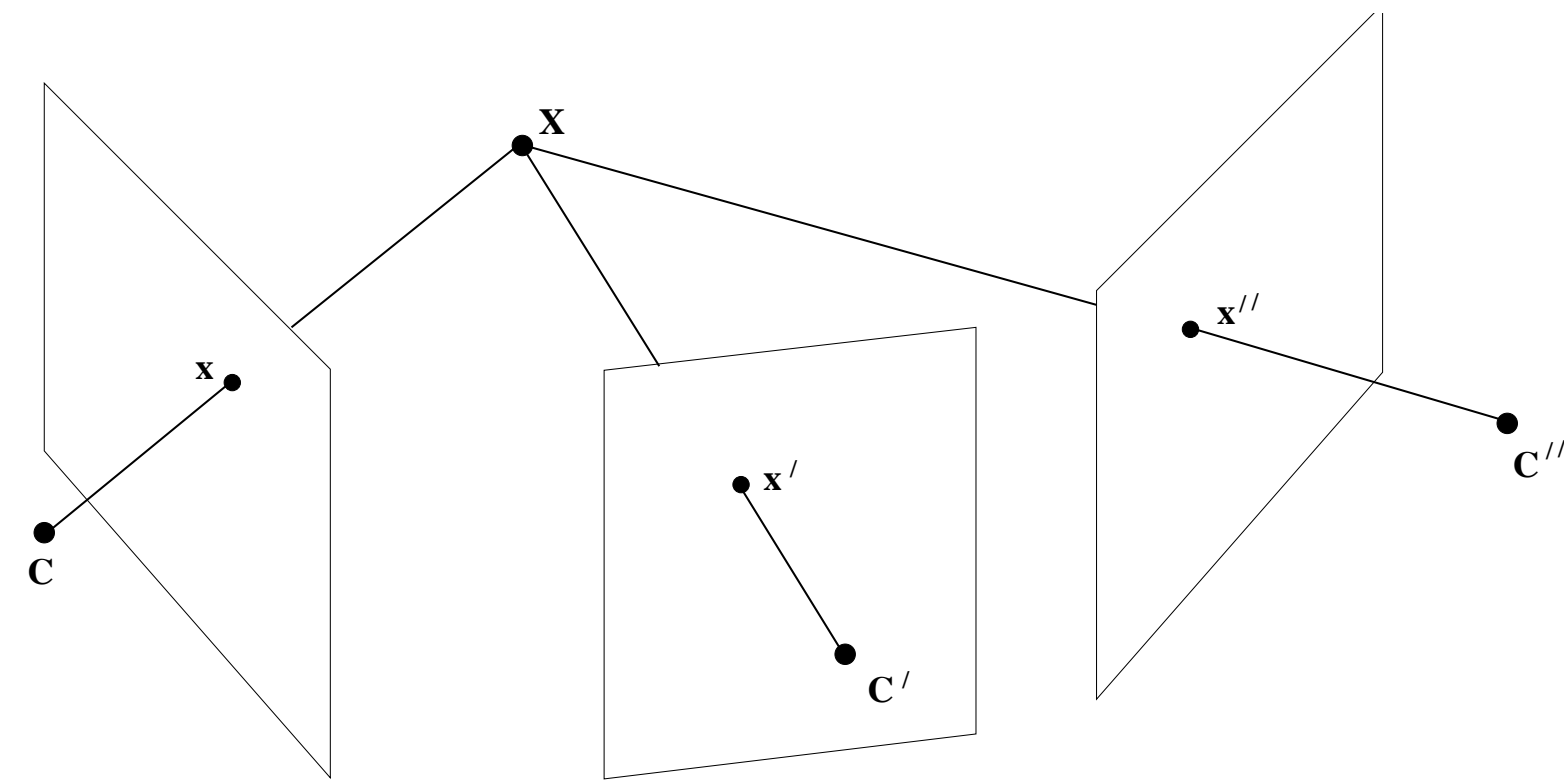
Has to be a zero matrix!

$$[\mathbf{x}']_{\times} \left(\sum_i x^i \mathbf{T}_i \right) [\mathbf{x}'']_{\times} = \mathbf{0}_{3 \times 3}$$

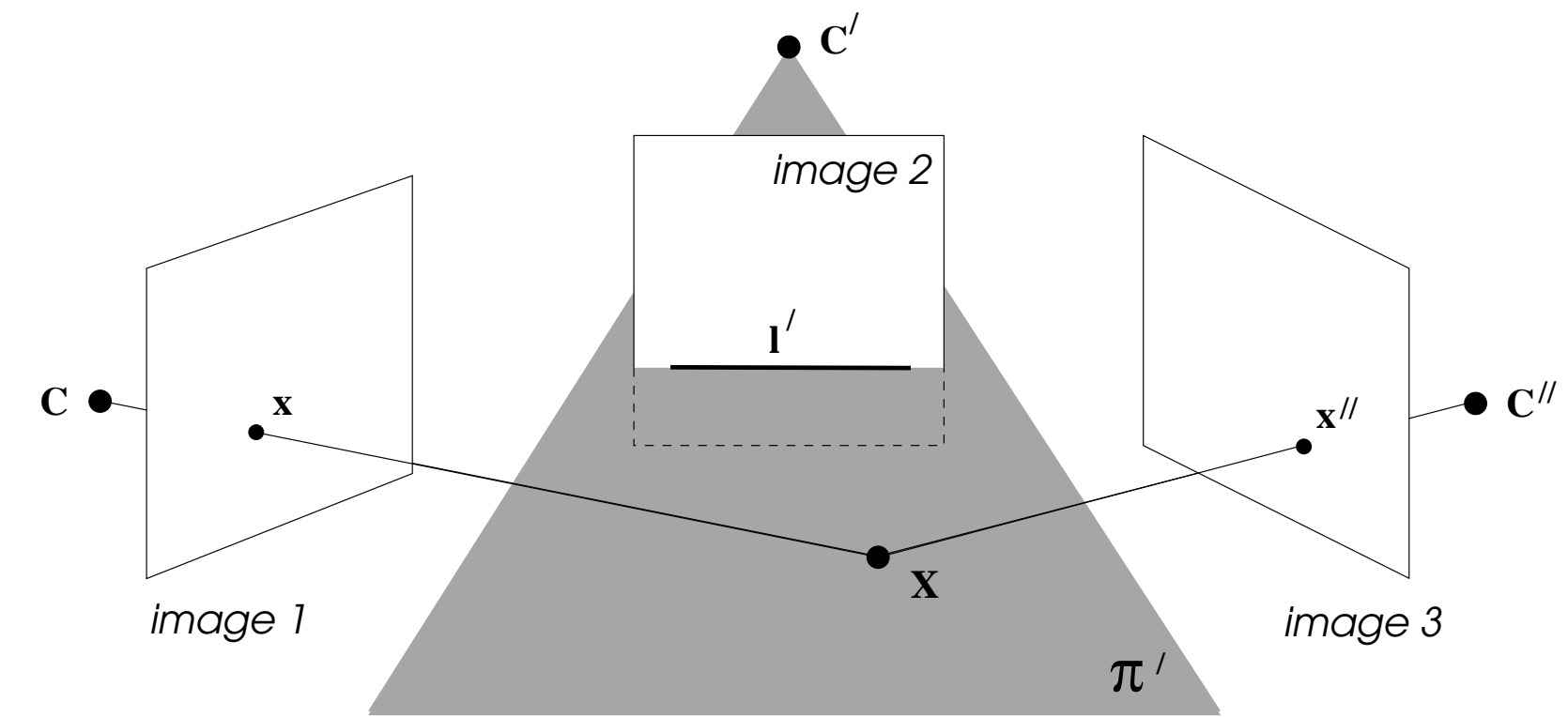
Summary

$$\mathbf{l}^\top = \mathbf{l}'^\top [\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3] \mathbf{l}''$$

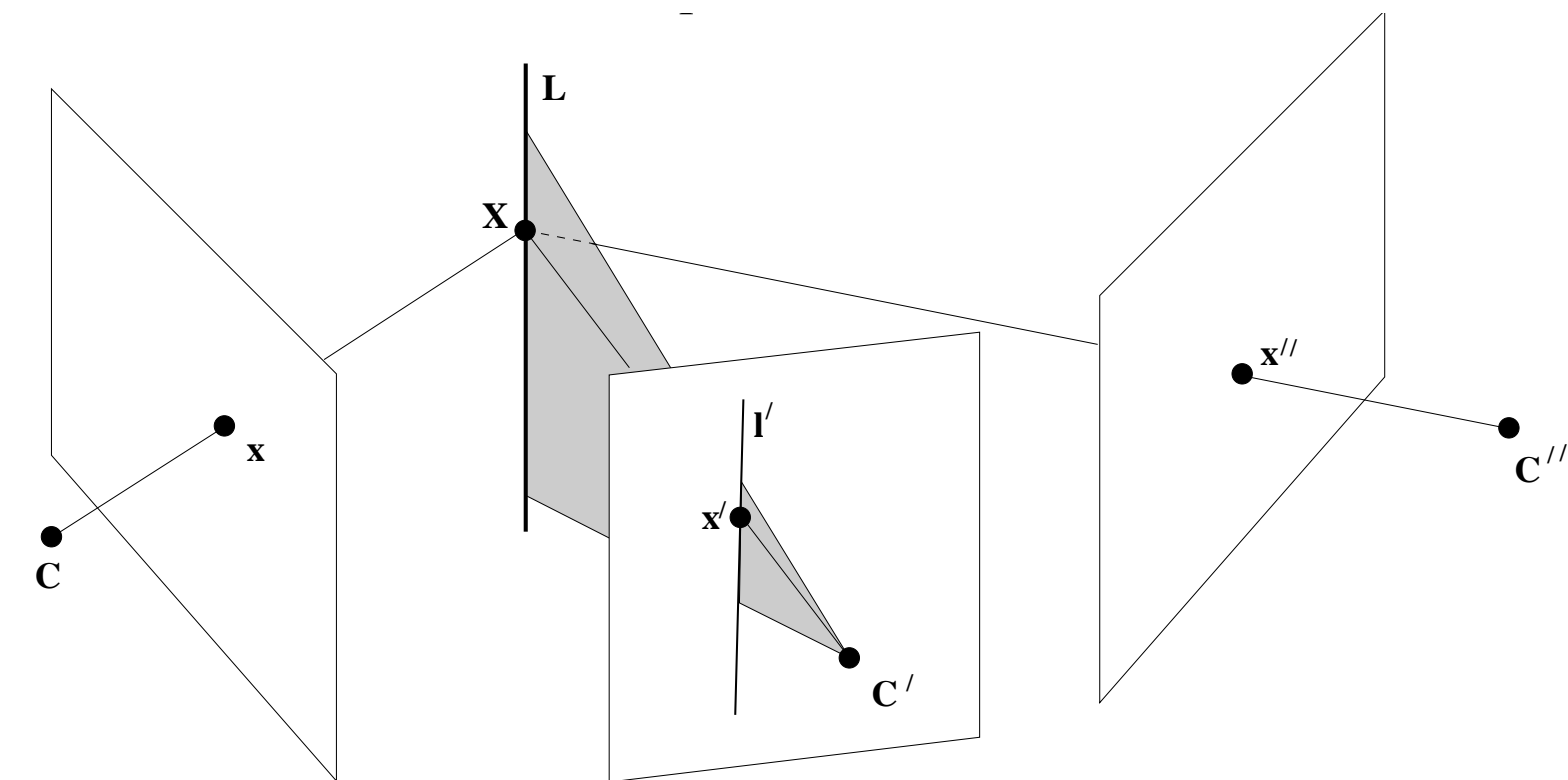
Tri-focal Tensor
(3 matrices of size 3 X 3)



$$[\mathbf{x}']_{\times} \left(\sum_i x^i \mathbf{T}_i \right) [\mathbf{x}'']_{\times} = \mathbf{0}_{3 \times 3}$$



$$\mathbf{H}_{13}(\mathbf{l}') = [\mathbf{T}_1^\top, \mathbf{T}_2^\top, \mathbf{T}_3^\top] \mathbf{l}'$$



$$\mathbf{l}'^\top \left(\sum_i x^i \mathbf{T}_i \right) [\mathbf{x}'']_{\times} = \mathbf{0}^\top$$

Questions?